

PROJECT 2 — ADJUSTABLE MECHANICAL ASSEMBLY

Project Type: CAD / 3D Modelling Capstone

Client Brief

Design an adjustable positioning mechanism consisting of a base, sliding carriage, mounting plate, guide block, adjustment screw, locking mechanism and associated hardware. The mounting plate must provide controlled linear adjustment.

Design Specification

- Adjustment travel: approximately 0–50 mm
- Minimum components: base, carriage, mounting plate, guide block, adjustment screw, locking knob, spacer and fasteners
- Provide 25, 50 and 75 mm travel variants through a common parametric design
- Demonstrate assembly motion and intended degrees of freedom

Parametric Requirements

TRAVEL, BASE_LENGTH, CARRIAGE_LENGTH, PLATE_THICKNESS, GUIDE_WIDTH, SCREW_PITCH, MOUNTING_PATTERN and other interface dimensions.

Required Skills

- Multi-part parametric modelling
- Assembly constraints/relationships
- Degrees of freedom
- Interference checking
- Clearance and fit
- Motion/kinematic reasoning
- Configuration/variant management
- Design intent communication

Client Approval Questions

- What is the functional motion?
- Which components are fixed and which are allowed to move?
- How are degrees of freedom controlled?
- Does the assembly contain interference?
- What happens when travel changes from 25 to 50 to 75 mm?
- Which commercial hardware is represented symbolically rather than modelled in full detail?

CAD QA / Validation

- Assembly constraint review
- DOF verification
- Interference/collision check

- Clearance check
- Variant regeneration
- Fastener/accessibility review
- STEP assembly export

Required Deliverables

- Native FreeCAD model (.FCStd)
- Manufacturing / engineering drawing (PDF)
- STEP export
- STL export where relevant
- Parameter/design-intent record
- CAD QA checklist/report